

$$x \times \frac{x}{5} = 0.008 \Rightarrow \frac{x^2}{5} = 0.008 \Rightarrow x^2 = 0.008 \times 5 = 0.04 \Rightarrow x = \sqrt{0.04} = 0.2.$$

$$\therefore \text{Greater number} = 0.2. \text{ Smaller number} = \frac{0.2}{5} = 0.04.$$

Hence, the required numbers are 0.2 and 0.04.

Ex. 17. Express as vulgar fractions:

(S.S.C., 2010)

(i) $0.\overline{37}$

(ii) $0.\overline{053}$

(iii) $0.\overline{001}$

(iv) $3.\overline{142857}$

Sol. (i) $0.\overline{37} = \frac{37}{99}$. (ii) $0.\overline{053} = \frac{53}{999}$. (iii) $0.\overline{001} = \frac{1}{999}$. (iv) $3.\overline{142857} = 3 + 0.\overline{142857} = 3 + \frac{142857}{999999} = 3\frac{142857}{999999}$.

Ex. 18. Express as vulgar fractions:

(i) $0.\overline{17}$

(ii) $0.12\overline{54}$

(iii) $2.5\overline{36}$

Sol. (i) $0.\overline{17} = \frac{17-1}{90} = \frac{16}{90} = \frac{8}{45}$. (ii) $0.12\overline{54} = \frac{1254-12}{9900} = \frac{1242}{9900} = \frac{69}{550}$.

(iii) $2.5\overline{36} = 2 + 0.5\overline{36} = 2 + \frac{536-53}{900} = 2 + \frac{483}{900} = 2 + \frac{161}{300} = 2\frac{161}{300}$.

Ex. 19. If $\frac{b}{a} = 0.25$, then what is the value of $\frac{2a-b}{2a+b} + \frac{2}{9}$?

(R.R.B., 2005)

Sol. $\frac{2a-b}{2a+b} + \frac{2}{9} = \frac{2-\frac{b}{a}}{2+\frac{b}{a}} + \frac{2}{9}$ [Dividing numerator and denominator by a]
 $= \frac{2-0.25}{2+0.25} + \frac{2}{9} = \frac{1.75}{2.25} + \frac{2}{9} = \frac{7}{9} + \frac{2}{9} = \frac{9}{9} = 1.$

Ex. 20. Find the value of $(1.99)^2$.

(Campus Recruitment, 2009)

Sol. $(1.99)^2 = (2 - 0.01)^2 = 2^2 + (0.01)^2 - 2 \times 2 \times 0.01 = 4 + 0.0001 - 0.04 = 3.9601.$

Ex. 21. Simplify: $\sqrt{\frac{0.009 \times 0.036 \times 0.016 \times 0.08}{0.002 \times 0.0008 \times 0.0002}}$

(S.S.C., 2010)

Sol. Since the sum of the decimal places in the numerator and denominator is the same, we have:

$$\text{Given expression} = \sqrt{\frac{9 \times 36 \times 16 \times 8}{2 \times 8 \times 2}} = \sqrt{1296} = 36.$$

Ex. 22. Simplify: $\frac{(0.35)^2 - (0.03)^2}{0.19}$.

(R.R.B., 2006)

Sol. Given expression = $\frac{(0.35+0.03)(0.35-0.03)}{0.19} = \frac{0.38 \times 0.32}{0.19} = 2 \times 0.32 = 0.64.$

Ex. 23. Find the value of $\frac{0.21 \times 0.21 \times 0.21 + 0.021 \times 0.021 \times 0.021}{0.63 \times 0.63 \times 0.63 + 0.063 \times 0.063 \times 0.063}$.

(S.S.C., 2006)

Sol. Let $a = 0.21$ and $b = 0.021$. Then,

$$\text{Given expression} = \frac{a^3 + b^3}{(3a)^3 + (3b)^3} = \frac{a^3 + b^3}{27a^3 + 27b^3} = \frac{a^3 + b^3}{27(a^3 + b^3)} = \frac{1}{27}.$$

Ex. 24. Simplify:

(i) $\left(\frac{2.75 \times 2.75 \times 2.75 - 2.25 \times 2.25 \times 2.25}{2.75 \times 2.75 + 2.75 \times 2.25 + 2.25 \times 2.25} \right)$

(C.P.O., 2007)

(ii) $\left(\frac{0.0347 \times 0.0347 \times 0.0347 + 0.9653 \times 0.9653 \times 0.9653}{0.0347 \times 0.0347 - 0.0347 \times 0.9653 + 0.9653 \times 0.9653} \right)$

(S.S.C., 2007)